Assignment 3

Q1. [8 Marks] For the given table titled **Characters**

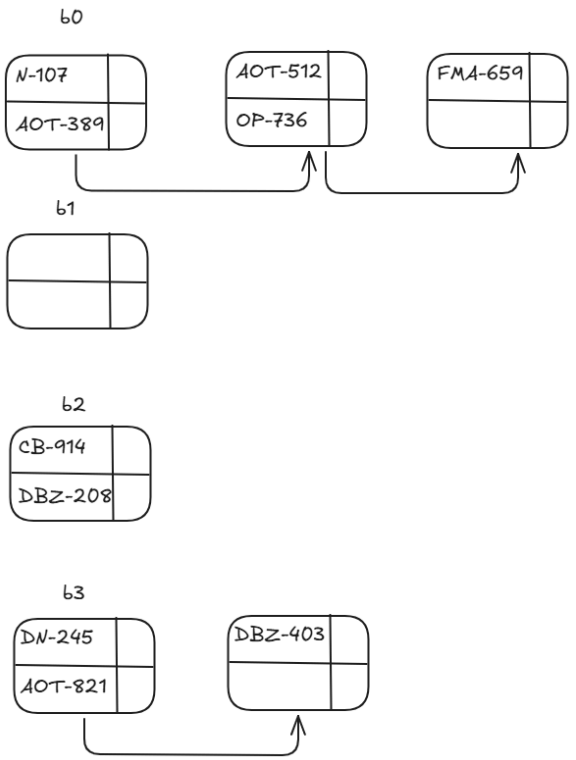
ID	Character	Anime	Role
N-107	Naruto Uzumaki	Naruto	Protagonist
DN-245	Light Yagami	Death Note	Protagonist
AOT-389	Mikasa Ackerman	Attack on Titan	Soldier
AOT-512	Levi Ackerman	Attack on Titan	Captain
OP-736	Monkey D. Luffy	One Piece	Pirate
AOT-821	Erren Yeager	Attack on Titan	Soldier
DBZ-403	Goku	Dragon Ball Z	Fighter
FMA-659	Edward Elric	Fullmetal Alchemist	Alchemist
CB-914	Spike Spiegel	Cowboy Bebop	Bounty Hunter
DBZ-208	Vegeta	Dragon Ball Z	Prince

Make a hash index on the attribute **ID**. Suppose the number of buckets in the hash index is 4 and each bucket can house 2 index entries at max. Bucket overflow is handled using forward chaining.

The hash function, h = (sum of all digits in ID) % 4
example: h(N-107) = (1+0+7) % 4 = 0
Only draw the buckets (with forward chaining, when applicable). No need to connect with the original table.

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h(N-107) = 1+0+7 = 8 % 4 = 0
h(DN-245) = 11 % 4 = 3
h(AOT-389) = 20 % 4 = 0
h(AOT-512) = 8 % 4 = 0
h(OP-736) = 16 % 4 = 0
h(AOT-821) = 11 % 4 = 3
h(DBZ-403) = 7 % 4 = 3
h(FMA-659) = 20 % 4 = 0
h(CB-914) = 14 % 4 = 2
h(DBZ-208) = 10 % 4 = 2



Q2. [12 Marks] A B+ Tree of order 3 is provided below.

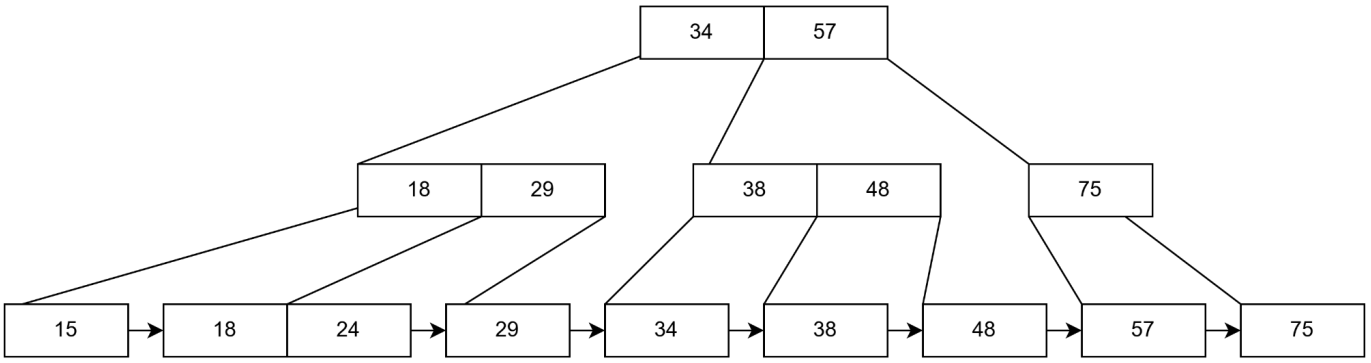


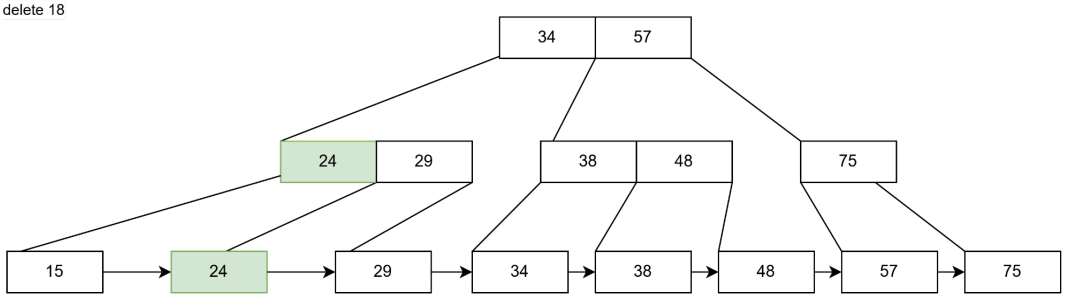
Figure: A

Do the following operations on the given tree (Figure: A) sequentially.

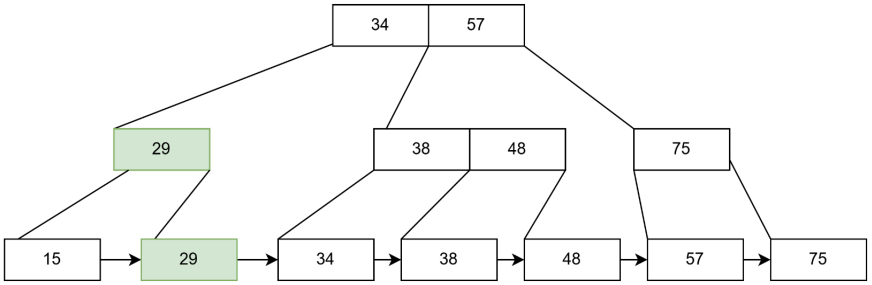
a. Delete 18
b. Delete 24
c. Delete 75
d. Delete 57
e. Delete 38
f. Delete 29

You need to draw the tree for each step after deletion.

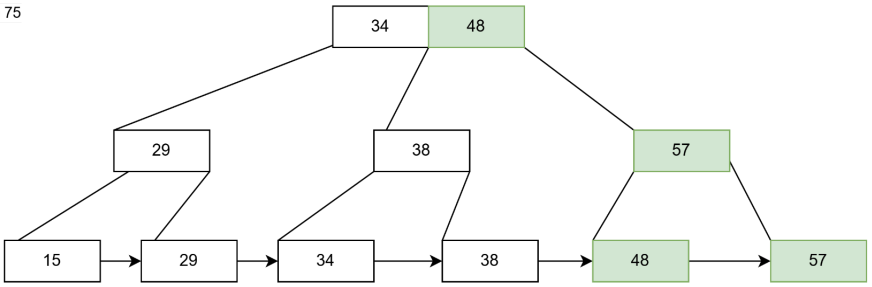
delete 18



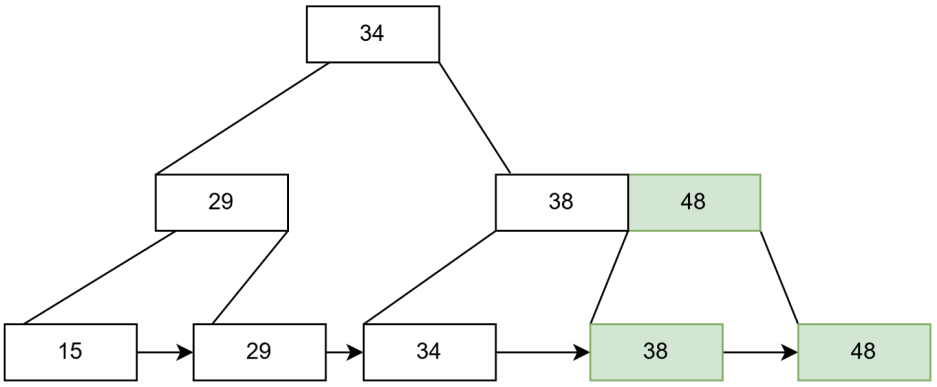
delete 24



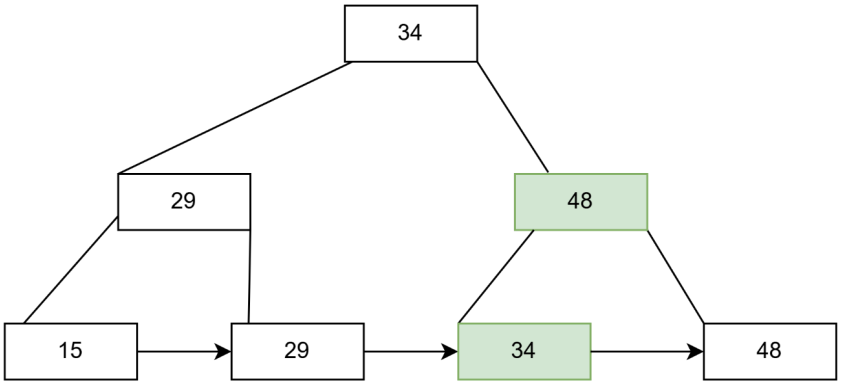
delete 75



delete 57



delete 38



delete 29

